



Faculty of Technology and Engineering

Chandubhai S. Patel Institute of Technology (CSPIT)

Department of Computer Science & Engineering

Date: 29/09/2026

Laboratory Manual

Academic Year	:	2026-27	Semester	:	3 rd
Course code	:	ITUE203	Course name	:	WEB DEVELOPMENT FRAMEWORKS

Practical – 4

JavaScript DOM Manipulation, Event Handling, and UI Interactivity

Problem Statement:- Add dynamic UI components such as collapsible FAQ, modal popup, image/content slider, notification banner, hamburger menu, and light/dark theme switcher using JavaScript.

Objective:- To design and develop a dynamic and interactive web page using JavaScript by implementing UI components such as a collapsible FAQ, modal popup, image/content slider, notification banner, hamburger menu, and light/dark theme switcher.

Github Repository Link:- <https://github.com/Nishtha-cse/StudentHubPortal-D26CS129>

Light/Dark theme switcher :

Theme.js file:

```
// ===== THEME SWITCHER =====

(function () {
  "use strict";
  var STORAGE_KEY = "studenthub-theme";
  var root = document.documentElement;

  function currentTheme() {
    return root.getAttribute("data-theme") === "dark" ? "dark" : "light";
  }

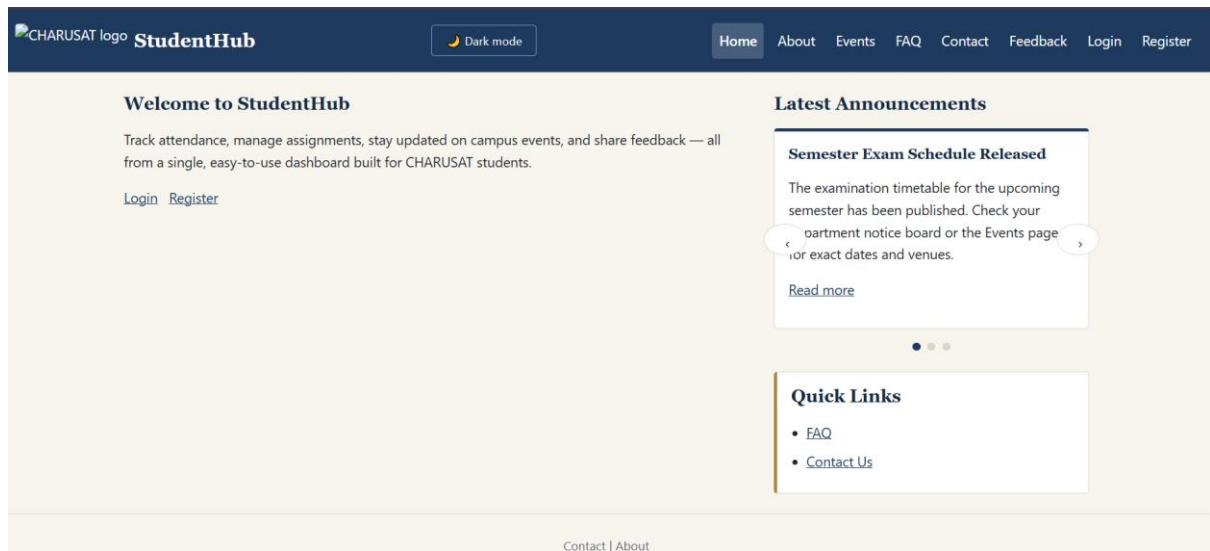
  function updateLabel(toggle) {
    var isDark = currentTheme() === "dark";
    toggle.setAttribute("aria-pressed", String(isDark));
    toggle.textContent = isDark ? "\u2600\uFE0F Light mode" : "\uD83C\uDF19
Dark mode";
  }

  document.addEventListener("DOMContentLoaded", function () {
    // Restore saved choice (the tiny script in <head> already avoids a flash)
    var stored = localStorage.getItem(STORAGE_KEY);
    if (stored === "dark" || stored === "light") {
      root.setAttribute("data-theme", stored);
    }

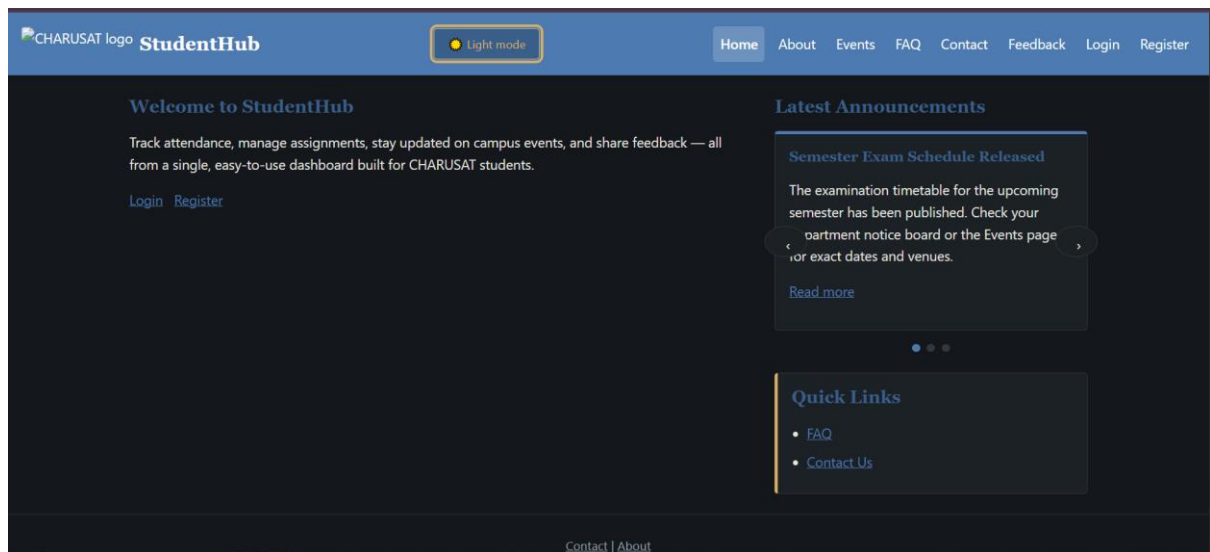
    var toggle = document.getElementById("theme-toggle");
    if (!toggle) return;
    updateLabel(toggle);

    toggle.addEventListener("click", function () {
      var next = currentTheme() === "dark" ? "light" : "dark";
      root.setAttribute("data-theme", next);
      localStorage.setItem(STORAGE_KEY, next);
      updateLabel(toggle);
    });
  });
})();
```

Home.html in light mode:



Home.html page in dark mode:



Collapsible FAQ :

Faqs.html :

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<link rel="stylesheet" href="../css/style.css">
<link rel="icon" href="../images/charusat.png" type="image/png">
<title>FAQs - StudentHub</title>
<meta name="description" content="Frequently asked questions about using StudentHub.">
<script>
  (function () {
    var saved = localStorage.getItem("studenthub-theme");
    if (saved === "dark" || saved === "light") {
      document.documentElement.setAttribute("data-theme", saved);
    }
  })();
</script>
</head>
<body>
<header>
  <div class="brand">
    
    <h1>Frequently Asked Questions</h1>
  </div>
  <div class="header-actions">
    <button type="button" id="theme-toggle" class="theme-toggle" aria-pressed="false">Dark mode</button>
    <button type="button" class="nav-toggle" aria-controls="primary-nav" aria-expanded="false">
      <span class="hamburger-icon" aria-hidden="true"></span>
      <span>Menu</span>
    </button>
  </div>
<nav aria-label="Main navigation" id="primary-nav">
  <ul>
    <li><a href="home.html">Home</a></li>
    <li><a href="about.html">About</a></li>
    <li><a href="event.html">Events</a></li>
    <li><a href="faqs.html" aria-current="page">FAQ</a></li>
    <li><a href="contact.html">Contact</a></li>
    <li><a href="feedback.html">Feedback</a></li>
    <li><a href="login.html">Login</a></li>
    <li><a href="register.html">Register</a></li>
  </ul>
</nav>
```

```
</header>

<main id="main-content">
  <section aria-label="Frequently asked questions">
    <div class="accordion-item">
      <h2>
        <button type="button" class="accordion-trigger" id="faq1-trigger"
aria-expanded="false" aria-controls="faq1-panel">
          How to reset the password?
        </button>
      </h2>
      <div class="accordion-panel" id="faq1-panel" role="region" aria-
labelledby="faq1-trigger" hidden>
        <p>You can reset your password by clicking the "Forgot Password" link
on the login page and following the instructions.</p>
      </div>
    </div>

    <div class="accordion-item">
      <h2>
        <button type="button" class="accordion-trigger" id="faq2-trigger"
aria-expanded="false" aria-controls="faq2-panel">
          How to update profile information?
        </button>
      </h2>
      <div class="accordion-panel" id="faq2-panel" role="region" aria-
labelledby="faq2-trigger" hidden>
        <p>You can update your profile information by navigating to the
"Profile" section and clicking the "Edit Profile" button.</p>
      </div>
    </div>

    <div class="accordion-item">
      <h2>
        <button type="button" class="accordion-trigger" id="faq3-trigger"
aria-expanded="false" aria-controls="faq3-panel">
          How to change the email address?
        </button>
      </h2>
      <div class="accordion-panel" id="faq3-panel" role="region" aria-
labelledby="faq3-trigger" hidden>
        <p>You can change your email address by going to the "Profile" section
and clicking the "Edit Profile" button.</p>
      </div>
    </div>
  </section>
</main>
```

```
<footer>
  <nav aria-label="Footer navigation">
    <a href="contact.html">Contact</a> |
    <a href="about.html">About</a>
  </nav>
  <p>&copy; 2026 StudentHub</p>
</footer>

<script src="../js/theme.js" defer></script>
<script src="../js/menu.js" defer></script>
<script src="../js/accordion.js" defer></script>
</body>
</html>
```

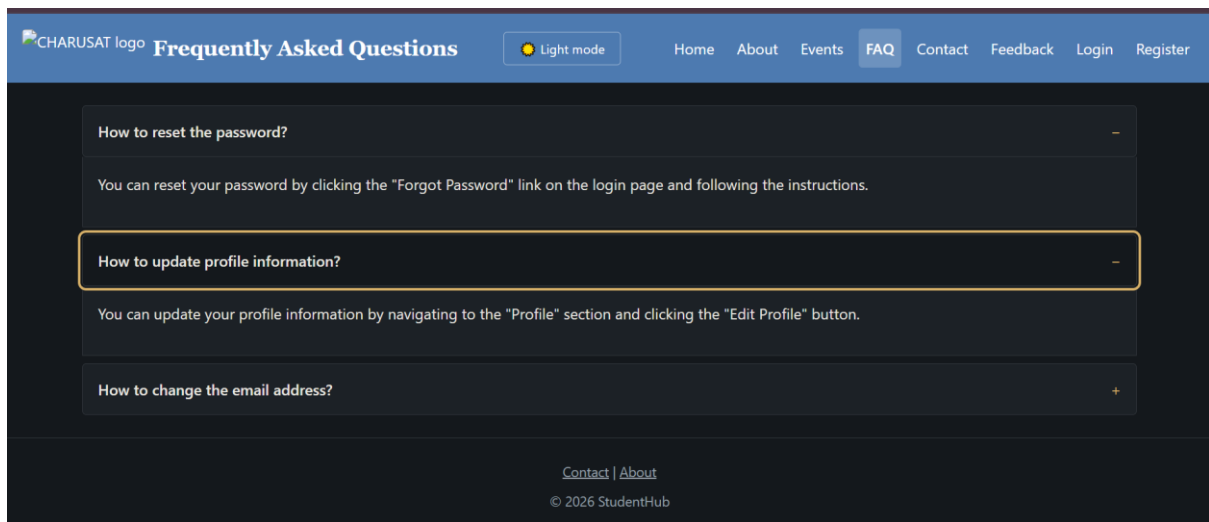
accordion.js :

<!-- ===== FAQ ACCORDION ===== -->

```
(function () {
  "use strict";

  document.addEventListener("DOMContentLoaded", function () {
    document.querySelectorAll(".accordion-trigger").forEach(function (trigger) {
      trigger.addEventListener("click", function () {
        var panelId = trigger.getAttribute("aria-controls");
        var panel = panelId ? document.getElementById(panelId) : null;
        var isExpanded = trigger.getAttribute("aria-expanded") === "true";

        trigger.setAttribute("aria-expanded", String(!isExpanded));
        if (panel) panel.hidden = isExpanded;
      });
    });
  });
})();
```



Modal popup

profile.html :

<!-- ===== MODAL POPUP ===== -->

```
<div class="modal-overlay" id="edit-profile-modal" role="dialog" aria-
modal="true" aria-labelledby="edit-profile-title" hidden>
  <div class="modal">
    <button type="button" class="modal-close" aria-label="Close
dialog">&times;</button>
    <h2 id="edit-profile-title">Edit Profile</h2>
    <form action="#" method="post">
      <label for="edit-phone">Phone Number</label><br>
      <input type="tel" id="edit-phone" name="phone" value="+91 98765
43210"><br><br>

      <label for="edit-semester">Semester</label><br>
      <input type="text" id="edit-semester" name="semester"
value="5th"><br><br>

      <button type="submit">Save Changes</button>
    </form>
  </div>
</div>
```

style.css :

/* ===== MODAL ===== */

```
.modal-overlay {
  position: fixed;
```

```

inset: 0;
background: rgba(20, 24, 28, 0.6);
display: flex;
align-items: center;
justify-content: center;
padding: var(--space-2);
z-index: 200;
}
.modal-overlay[hidden] { display: none; }
.modal {
  background: var(--color-surface);
  border-radius: var(--radius);
  padding: var(--space-3);
  max-width: 28rem;
  width: 100%;
  position: relative;
}
.modal-close {
  position: absolute;
  top: var(--space-1);
  right: var(--space-1);
  background: transparent;
  border: none;
  padding: 0;
  font-size: 1.3rem;
  cursor: pointer;
  color: var(--color-muted);
}

```

modal.js :

```
// ===== MODAL POPUP =====
```

```

(function () {
  "use strict";

  document.addEventListener("DOMContentLoaded", function () {
    var lastTrigger = null;

    function openModal(modal, trigger) {
      lastTrigger = trigger;
      modal.hidden = false;
      document.body.classList.add("modal-open");
      var focusTarget = modal.querySelector("input, textarea, select, button, [href]");
      if (focusTarget) focusTarget.focus();
    }

    function closeModal(modal) {
      modal.hidden = true;
    }
  });
}

```



```
document.body.classList.remove("modal-open");
if (lastTrigger) lastTrigger.focus();
}

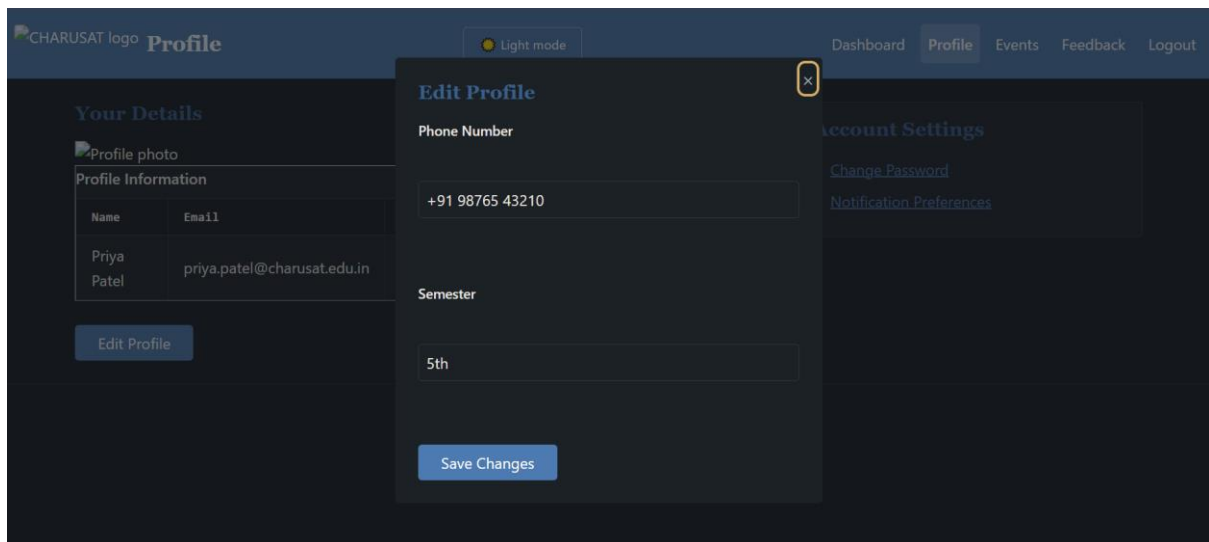
// Any button with data-modal-target="<modal id>" opens that modal
document.querySelectorAll("[data-modal-target]").forEach(function
(trigger) {
    var modal = document.getElementById(trigger.dataset.modalTarget);
    if (!modal) return;
    trigger.addEventListener("click", function () { openModal(modal,
trigger); });
});

document.querySelectorAll(".modal-overlay").forEach(function (modal) {
    var closeBtn = modal.querySelector(".modal-close");
    if (closeBtn) closeBtn.addEventListener("click", function () {
closeModal(modal); });

    // Click on the dark background closes it
    modal.addEventListener("click", function (event) {
        if (event.target === modal) closeModal(modal);
    });

    modal.addEventListener("keydown", function (event) {
        if (event.key === "Escape") { closeModal(modal); return; }

        // Keep Tab focus inside the open dialog
        if (event.key === "Tab") {
            var items = modal.querySelectorAll("button, [href], input, textarea,
select");
            if (!items.length) return;
            var first = items[0];
            var last = items[items.length - 1];
            if (event.shiftKey && document.activeElement === first) {
                event.preventDefault(); last.focus();
            } else if (!event.shiftKey && document.activeElement === last) {
                event.preventDefault(); first.focus();
            }
        }
    });
});
});
})();
```



Image/content slider :

home.html:

<!-- ===== CONTENT SLIDER ===== -->

```
<div class="slider" aria-roledescription="carousel" aria-label="Latest
announcements">
  <div class="slider-track">
    <article class="slide">
      <h3>Semester Exam Schedule Released</h3>
      <p>The examination timetable for the upcoming semester has been
published. Check your department notice board or the Events page for exact
dates and venues.</p>
      <p><a href="event.html">Read more</a></p>
    </article>

    <article class="slide" hidden>
      <h3>Annual Tech Fest Registrations Open</h3>
      <p>Sign up now for workshops, coding contests, and guest talks at
this year's tech fest. Limited seats available for select sessions.</p>
      <p><a href="event.html">Read more</a></p>
    </article>

    <article class="slide" hidden>
      <h3>Library Timings Extended</h3>
      <p>The central library will now stay open until 9 PM on weekdays
during the exam period to support student preparation.</p>
      <p><a href="event.html">Read more</a></p>
    </article>
  </div>
</div>
```

```

        <button type="button" class="slider-prev" aria-label="Previous
announcement">&#8249;</button>
        <button type="button" class="slider-next" aria-label="Next
announcement">&#8250;</button>
        <div class="slider-dots" role="tablist" aria-label="Choose
announcement"></div>

</div>

```

style.css:

```
/* ===== CONTENT SLIDER ===== */
```

```

.slider { position: relative; }
.slider-track { position: relative; min-height: 8rem; }
.slide[hidden] { display: none; }
.slider-prev, .slider-next {
  position: absolute;
  top: 50%;
  transform: translateY(-50%);
  background: var(--color-surface);
  border: 1px solid var(--color-border);
  border-radius: 50%;
  width: 2.25rem;
  height: 2.25rem;
  cursor: pointer;
  font-size: 1.1rem;
}
.slider-prev { left: -0.75rem; }
.slider-next { right: -0.75rem; }
.slider-dots {
  display: flex;
  justify-content: center;
  gap: var(--space-1);
  margin-top: var(--space-2);
}
.slider-dots button {
  width: 0.6rem;
  height: 0.6rem;
  border-radius: 50%;
  border: none;
  background: var(--color-border);
  padding: 0;
  cursor: pointer;
}
.slider-dots button[aria-selected="true"] { background: var(--color-primary); }

```

Slider.js:

```
// ===== CONTENT SLIDER =====
```

```
/* Feature 6: Content slider (prev / next buttons, dots, keyboard arrows) */
(function () {
    "use strict";

    document.addEventListener("DOMContentLoaded", function () {
        document.querySelectorAll(".slider").forEach(function (slider) {
            var slides = Array.from(slider.querySelectorAll(".slide"));
            if (slides.length === 0) return;

            var prevBtn = slider.querySelector(".slider-prev");
            var nextBtn = slider.querySelector(".slider-next");
            var dotsWrap = slider.querySelector(".slider-dots");
            var index = 0;

            // Build one dot per slide
            if (dotsWrap) {
                dotsWrap.innerHTML = "";
                slides.forEach(function (_, i) {
                    var dot = document.createElement("button");
                    dot.type = "button";
                    dot.setAttribute("role", "tab");
                    dot.setAttribute("aria-label", "Go to slide " + (i + 1));
                    dot.addEventListener("click", function () { show(i); });
                    dotsWrap.appendChild(dot);
                });
            }
            var dots = dotsWrap ? Array.from(dotsWrap.children) : [];

            function show(newIndex) {
                index = (newIndex + slides.length) % slides.length; // wraps around
                slides.forEach(function (slide, i) { slide.hidden = i !== index; });
                dots.forEach(function (dot, i) {
                    dot.setAttribute("aria-selected", String(i === index));
                });
            }

            if (prevBtn) prevBtn.addEventListener("click", function () { show(index - 1); });
            if (nextBtn) nextBtn.addEventListener("click", function () { show(index + 1); });

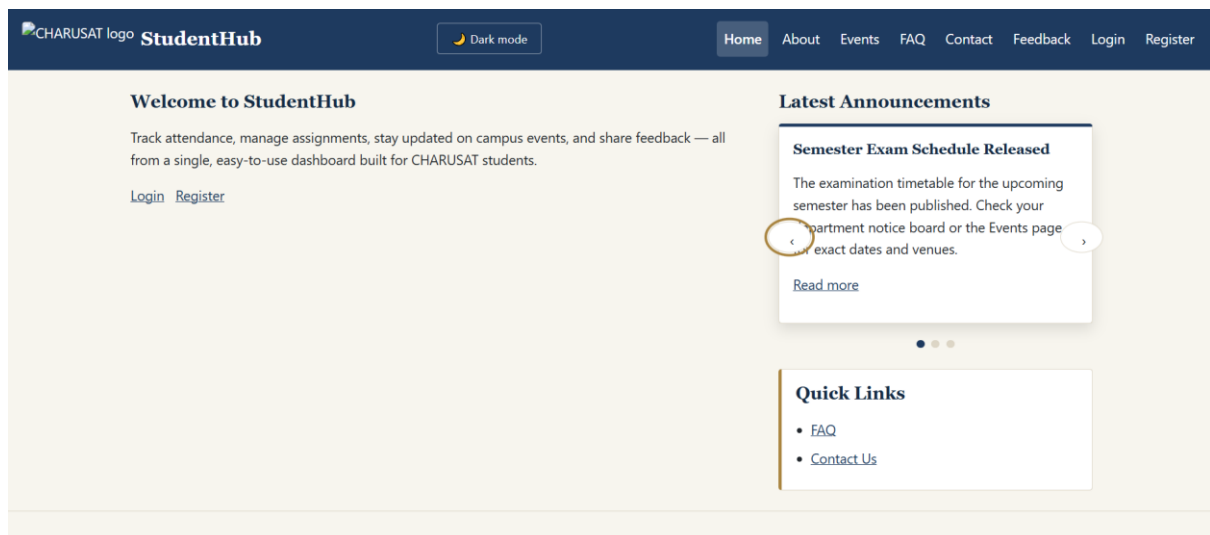
            slider.addEventListener("keydown", function (event) {
                if (event.key === "ArrowLeft") show(index - 1);
            });
        });
    });
})();
```

```

        if (event.key === "ArrowRight") show(index + 1);
    });

    show(0);
  });
});
})();

```



Notification banner :

home.html:

```
<!-- ===== NOTIFICATION BANNER ===== -->
```

```

<div class="notice-banner" id="notice-banner" data-notice-id="techfest-2026"
role="status">
  <p>New: Annual Tech Fest registrations are now open &mdash; see Latest
Announcements below.</p>
  <button type="button" class="notice-close" aria-label="Dismiss
notification">&times;</button>
</div>

```

Style.css:

```
/* ===== NOTIFICATION BANNER ===== */
```

```

.notice-banner {
  background: var(--color-accent);
  color: var(--color-primary-dark);
  padding: var(--space-1) var(--space-2);
}

```

```

display: flex;
align-items: center;
justify-content: center;
gap: var(--space-2);
font-size: 0.9rem;
text-align: center;
}
.notice-banner[hidden] { display: none; }
.notice-banner p { margin: 0; }
.notice-close {
background: transparent;
border: none;
padding: 0;
font-size: 1.2rem;
line-height: 1;
cursor: pointer;
color: inherit;
}

```

banner.js:

```
// ===== NOTIFICATION BANNER =====
```

```

/* Feature 3: Notification banner (dismissal remembered with localStorage) */
(function () {
  "use strict";

  document.addEventListener("DOMContentLoaded", function () {
    var banner = document.getElementById("notice-banner");
    if (!banner) return;

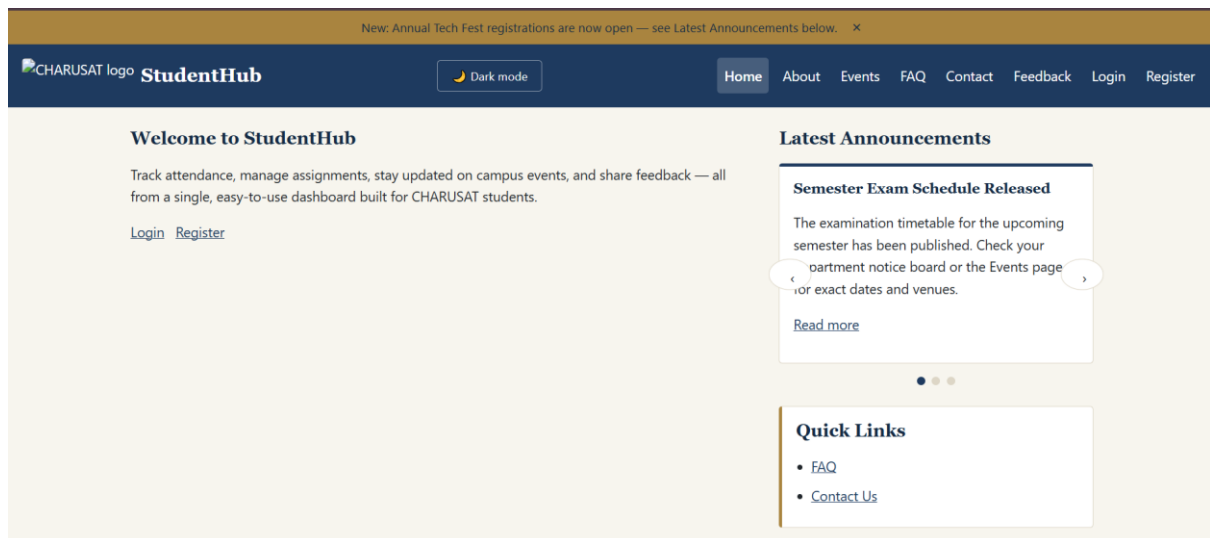
    var noticeId = banner.dataset.noticeId || "default";
    var storageKey = "studenthub-notice-" + noticeId;

    if (localStorage.getItem(storageKey) === "dismissed") {
      banner.hidden = true;
      return;
    }

    var closeBtn = banner.querySelector(".notice-close");
    if (!closeBtn) return;

    closeBtn.addEventListener("click", function () {
      banner.hidden = true;
      localStorage.setItem(storageKey, "dismissed");
    });
  });
})();

```



Hamburger menu

Login.html:

```
<div class="header-actions">
  <button type="button" id="theme-toggle" class="theme-toggle" aria-
pressed="false">Dark mode</button>
  <button type="button" class="nav-toggle" aria-controls="primary-nav" aria-
expanded="false">
    <span class="hamburger-icon" aria-hidden="true"></span>
    <span>Menu</span>
  </button>
</div>
```

 Dark mode

 Menu

Home

About

Events

FAQ

Contact

Feedback

Login

Register

Login to Your Account

Username

Password

Login

[Forgot Password?](#)